Group Name & Number	Group 40
Course	Electronic Product Design & Reverse Engineering (EPDRE)

Reverse Engineering of the Logitech M150 Wireless Mouse

EPDRE Project Report | 102 × 62 × 37 mm, 80 g

1. Title of Project

Reverse Engineering Activity Report: Logitech M150 Wireless Mouse.

2. Objective of Project

To study the Logitech M150 through external observation and teardown, identify its major electronic and mechanical components, and explain how optical sensing, embedded control, wireless communication and power management work together.

3. Components and Tools Used

Tools: the M150 mouse and USB receiver, teardown photographs, a computer and Microsoft Word for documentation. No electrical measurements or firmware extraction were performed.

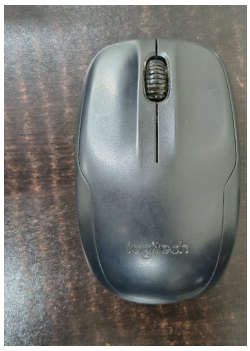


Fig. 1: M150 before teardown



Fig. 2: Teardown (base + PCB, top shell)

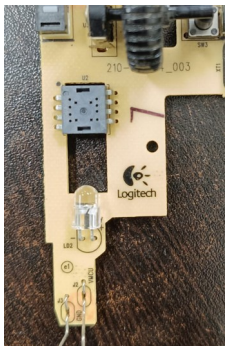


Fig. 3: PCB: sensor U2, LED LD2, SW3

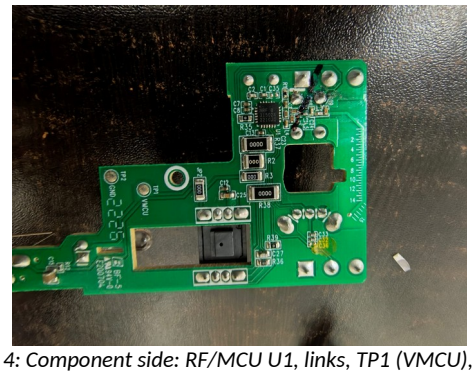


Fig. 4: Component side: RF/MCU U1, links, TP1 (VMCU), TP2 (GND)

Ref	Component	Function	Specification / marking
U1	RF transceiver + MCU IC	Processes inputs, debounces switches, builds and sends 2.4 GHz packets	20-pin QFN, "NRF 31502A 2210JQ" (Nordic nRF-family; date code wk 10, 2022)
U2	Optical navigation sensor	Images the surface to track X/Y movement	8-pin, likely PixArt-type; exact part no. not read
LD2	Illumination LED	Lights the surface for the sensor	Ratings not measured
XT1	Crystal	Timing reference	Value not read
SW3 +	Micro-switches, scroll wheel/encoder	Click and scroll inputs	Mechanical; firmware debounce needed
R2, R3, R37, R38, JP2	Zero-ohm links	Configuration jumpers for design variants	Marked "000" / "0000"
C1, C2, C7, C8, C12, C13, C25, C27, C33...	Ceramic SMD capacitors	Power decoupling and filtering	Values not read
TP1, TP2	Test points	Probe supply rail (VMCU) and ground	–

4. Circuit Diagram (Functional Block Diagram)

A verified electrical schematic was not available from the teardown, so a functional block diagram is shown instead of presenting an unverified circuit as exact.

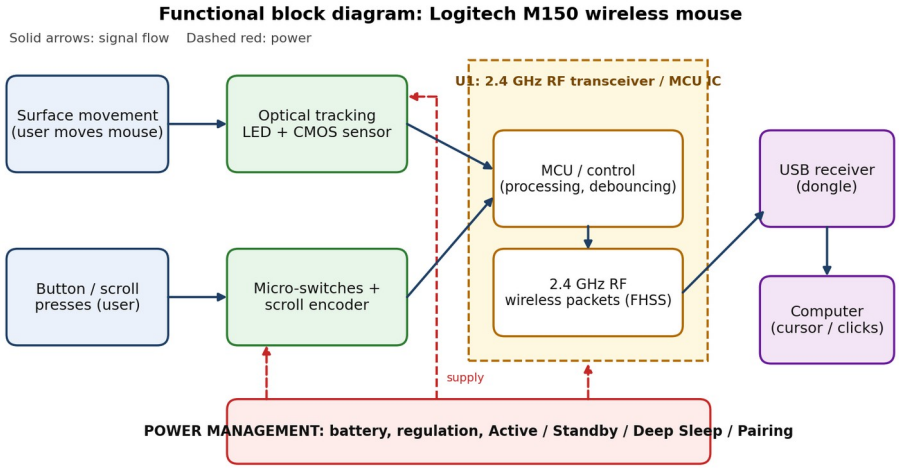


Fig. 5: Functional block diagram of the mouse

5. Explanation

5.1 External features and purpose

The M150 converts hand movement and button/scroll actions into digital data sent wirelessly through a small USB receiver, which moves the pointer and issues clicks and scrolls on the computer. Even before opening it, it can be seen as a small embedded system: mechanical input, sensing, processing, communication and power management in a 102 × 62 × 37 mm shell.

5.2 Modes of operation

Mode	Description
Active	Full power for real-time tracking and click transmission
Standby	After short stillness, sensor activity is reduced but response stays quick
Deep sleep	After longer inactivity most functions sleep; movement or input wakes the mouse
Pairing	Wireless subsystem active while establishing a link with the USB receiver

Why it matters: aggressive sleep saves energy but increases wake-up delay; staying active gives readiness but shortens battery life.

5.3 Subsystems

Subsystem	Function	Working principle	Characteristics / role
Optical tracking	Detects movement over the surface	LED lights the surface; a lens focuses it on a CMOS sensor; change between successive images gives X/Y movement	No moving parts, but struggles on glossy/transparent glass; replaces ball and rollers
MCU / control	Collects and prepares data	Processes sensor, switch and scroll data; firmware debounces switches; passes clean digital data to RF block	Low power; manages Active/Standby/Deep Sleep; coordinates all blocks
2.4 GHz RF + USB receiver	Sends data without a cable	Data becomes wireless packets, sent with frequency hopping (FHSS) in the 2.4 GHz ISM band to the USB receiver	No line-of-sight needed; pairing mode links mouse and receiver
Power management	Powers circuit, extends battery life	Battery energy regulated and distributed; timing moves the mouse between power states; movement or input wakes it	Trade-off between wake-up delay and energy saving
Mechanical input	Clicks and scrolling	Micro-switch press or wheel turn gives electrical signals read by the MCU	Contact bounce must be filtered

5.4 Complete working flow

1. The user moves the mouse or presses/scrolls a control.
2. LED and CMOS sensor capture surface images; micro-switches and scroll encoder sense inputs.
3. The MCU processes the data and debounces switches into clean digital data.
4. The RF block sends packets, hopping channels in the 2.4 GHz band to avoid interference.

5. The USB receiver passes the data to the computer, which moves the cursor or performs the click/scroll.
6. Throughout, power management regulates the battery supply and switches between Active, Standby and Deep Sleep.

Signal chain: movement / button input → optical and mechanical sensing → MCU → wireless packet → USB receiver → computer.

5.5 Evolution of the mouse and design trade-offs

Stage	Improvement	Drawback
Mechanical ball mice	Simple X/Y coordinates via rollers	Wear and contamination
Early optical / IR wireless	Fewer moving parts; no cable	Surface limits, line-of-sight, lag
2.4 GHz RF	No line-of-sight dependence; works on ordinary opaque surfaces	Radio interference
Modern office mice (M150)	Small receivers, better sensing, automatic power states	Still struggles on glossy/transparent glass

Key trade-offs: power vs performance; wireless mobility vs interference; optical accuracy vs surface compatibility; size vs integration. Smart features used in the M150: intelligent power saving, adaptive frequency hopping (FHSS) and switch debouncing.

5.6 Conclusion

The M150 is a compact wireless optical mouse built from optical tracking, MCU processing, 2.4 GHz RF and power management. Reverse engineering is not just identifying parts; it is understanding the engineering decisions that tie sensing, embedded processing, wireless communication and power management into one working product. Exact IC ratings need datasheets or measurements and are left open rather than guessed.

6. Simulation Circuit

No TinkerCAD/KiCAD simulation is included. The M150 uses a proprietary integrated RF/MCU design, and its verified schematic, IC datasheets and firmware are not available from a visual teardown. Fig. 5 explains operation but is not a simulated electrical circuit.

7. Code

The original firmware is embedded in the controller and was not extracted, so no product code is provided or claimed as part of this study.

TinkerCAD / KiCAD simulation link (GitHub): _____